LAB 7 | Dynamic Database in Prolog

Assignment: Library Book Management System Using Dynamic Predicates

Assignment Title

Library Book Management Using Dynamic Predicates in Prolog

Objective

Develop a Library Book Management System using Prolog's dynamic database features. Learn to add, update, delete, search, and display information during program execution.

- Add new books to the library.
- Update the availability status of a book.
- Remove books from the library.
- Display all stored book records.
- Search for a book by title.
- Exit the system.

Task Description

Write a Prolog program that manages library book records using a dynamic predicate. Each record follows this structure:

```
book(Title, Author, Status).
```

Title is the book name, Author is the author, and Status is either available or borrowed.

Example initial records:

```
:- dynamic book/3.

book(prolog_basics, john, available).
book(artificial_intelligence, peter, available).
book(data_structures, david, borrowed).
```

Required Operations

Operation	Description
Add Book	Insert a new book record using assertz/1
Update Book Status	Change status using retract/1 and assertz/1
Remove Book	Delete an existing record using retract/1
Show All Books	Display every book in the database
Search Book	Find a book by its title
Exit	Terminate the program

Program Requirements

The program must accept interactive input using read/1 and display this menu:

```
===== LIBRARY BOOK MANAGEMENT SYSTEM =====
1. Add New Book
2. Update Book Status
3. Remove Book
4. Show All Books
5. Search Book
6. Exit
Enter your choice (1-6):
```

Sample Execution

Step 1: Add a new book

```
Enter your choice (1-6): 1.
Enter book title: machine_learning.
Enter author name: alex.
Enter book status: available.
Book added successfully.
```

Step 2: Display all books

```
Enter your choice (1-6): 4.
Library Book Records:
Title: prolog_basics          Author: john    Status: available
Title: artificial_intelligence Author: peter Status: available
Title: data_structures       Author: david   Status: borrowed
Title: machine_learning      Author: alex    Status: available
```

Step 3: Update book status

```
Enter your choice (1-6): 2.
Enter book title: machine_learning.
Enter new status: borrowed.
Book status updated successfully.
```

Step 4: Search for a book

```
Enter your choice (1-6): 5.
Enter book title: machine_learning.
Book Found!
Title: machine_learning
Author: alex
Status: borrowed
```

Step 5: Remove a book

```
Enter your choice (1-6): 3.
Enter book title: machine_learning.
Book removed successfully.
```

Step 6: Exit the system

```
Enter your choice (1-6): 6.
Thank you for using the Library Management System!
```

Instructions for Students

- Declare book/3 as a dynamic predicate.
- Use assertz/1 to add records and retract/1 to delete them.
- Use retract/1 and assertz/1 together to update a record.
- Use read/1 for user input and implement a menu-driven interface.
- Show an appropriate message if the requested book is not found.
- Prevent duplicate book titles from being added.
- Test all operations in SWI-Prolog.

Expected Learning Outcomes

- Understand and declare dynamic predicates in Prolog.
- Insert, remove, and update facts at runtime.
- Build interactive menu-driven applications.
- Apply dynamic databases to a practical record-management problem.

Submission Requirements

Submit a complete Prolog source file (.pl), screenshots showing successful execution of each operation, and a brief explanation of how the dynamic database changes during execution.

Note: Students must implement the program independently using the concepts covered in Lab 7.